HOPE LAB / COMPUTATIONAL WORKFLOW

16S rRNA Amplicon Analysis with QIIME 2 and DADA2

A reproducible computational workflow for demultiplexed FASTQ data; commands require dataset-specific parameter selection.

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Scope and expected inputs

This workflow begins with demultiplexed 16S rRNA amplicon FASTQ files and a sample metadata table. It covers data import, sequence-quality inspection, DADA2 denoising, taxonomy assignment and diversity analysis. It does not cover wet-lab sampling, DNA extraction, PCR, library preparation or sequencing.

1. Prepare the project files

- Keep raw FASTQ files read-only. Work in a separate analysis directory and record checksums or file sizes.
- Create a sample metadata TSV with one unique sample ID per row. Include relevant study variables and control labels; do not place spaces or duplicated IDs in sample identifiers.
- Create a QIIME 2 FASTQ manifest with sample IDs and absolute file paths. Use the manifest format that matches single-end or paired-end reads and the Phred encoding supplied by the sequencing facility.
- Confirm that each sample ID in the manifest matches the metadata table and that all input files exist.

2. Import demultiplexed paired-end FASTQ data

```
qiime tools import \
  --type 'SampleData[PairedEndSequencesWithQuality]' \
  --input-path manifest.tsv \
  --output-path demux-paired-end.qza \
  --input-format PairedEndFastqManifestPhred33V2
```

Use the corresponding single-end semantic type and manifest format for single-end data. Confirm the FASTQ quality encoding with the sequencing provider; do not assume Phred33 if the provider specifies otherwise.

3. Inspect the read-quality profiles

```
qiime demux summarize \
  --i-data demux-paired-end.qza \
  --o-visualization demux-summary.qzv

qiime tools view demux-summary.qzv
```

- Review per-cycle quality profiles, read counts, length distribution and sample balance.
- Choose trimming and truncation values from the actual quality profile and amplicon design. For paired-end reads, retain enough high-quality overlap to merge the forward and reverse reads.
- There is no universal truncation length. Record the rationale for every selected parameter.

4. Remove primers only when they remain in the reads

```
qiime cutadapt trim-paired \
  --i-demultiplexed-sequences demux-paired-end.qza \
  --p-front-f FORWARD_PRIMER_SEQUENCE \
  --p-front-r REVERSE_PRIMER_SEQUENCE \
  --o-trimmed-sequences demux-trimmed.qza
```

Replace the primer placeholders with the actual primer sequences and confirm orientation against the sequencing design. If the sequencing facility has already removed primers, do not trim them a second time. Review cutadapt summaries before denoising.

5. Denoise paired-end reads with DADA2

```
qiime dada2 denoise-paired \
  --i-demultiplexed-seqs demux-trimmed.qza \
  --p-trim-left-f TRIM_LEFT_F \
  --p-trim-left-r TRIM_LEFT_R \
  --p-trunc-len-f TRUNC_LEN_F \
  --p-trunc-len-r TRUNC_LEN_R \
  --o-table table.qza \
  --o-representative-sequences rep-seqs.qza \
  --o-denoising-stats denoising-stats.qza \
  --o-base-transition-stats base-transition-stats.qza
```

- Replace the four parameter placeholders with integer values selected from the quality plots and primer design. The values above are intentionally not pre-filled because they depend on the sequencing run and target region.
- For single-end reads, use the q2-dada2 denoise-single action and its single-end parameter names.
- Inspect the denoising statistics and base-transition diagnostics. Review input reads, filtered reads, denoised reads, merged reads and non-chimeric reads by sample.

6. Review the feature table and representative sequences

```
qiime metadata tabulate \
  --m-input-file denoising-stats.qza \
  --o-visualization denoising-stats.qzv

qiime feature-table summarize \
  --i-table table.qza \
  --m-sample-metadata-file metadata.tsv \
  --o-visualization table-summary.qzv

qiime feature-table tabulate-seqs \
  --i-data rep-seqs.qza \
  --o-visualization rep-seqs.qzv
```

- Check sample retention, feature frequency distribution, sequence lengths, and whether negative controls contain non-trivial features.
- Do not apply a fixed prevalence or frequency filter without a documented rationale. Keep a record of all exclusions.

7. Assign taxonomy with a region-appropriate classifier

```
qiime feature-classifier classify-sklearn \
  --i-classifier classifier-matched-to-region.qza \
  --i-reads rep-seqs.qza \
  --o-classification taxonomy.qza

qiime metadata tabulate \
  --m-input-file taxonomy.qza \
  --o-visualization taxonomy.qzv
```

- Use a classifier built from a versioned reference database and matching the amplified region and primer design. Record the database release, classifier training details and QIIME 2 version.
- Taxonomic assignments are limited by marker-gene resolution and database coverage; avoid claiming species-level resolution unless supported.

8. Visualize taxonomy and community structure

```
qiime taxa barplot \
  --i-table table.qza \
  --i-taxonomy taxonomy.qza \
  --m-metadata-file metadata.tsv \
  --o-visualization taxa-barplot.qzv
```

- Review the composition across samples and groups. Treat relative-abundance plots as compositional summaries, not absolute cell counts.

9. Phylogeny and diversity (when appropriate)

```
qiime phylogeny align-to-tree-mafft-fasttree \
  --i-sequences rep-seqs.qza \
  --o-alignment aligned-rep-seqs.qza \
  --o-masked-alignment masked-aligned-rep-seqs.qza \
  --o-tree unrooted-tree.qza \
  --o-rooted-tree rooted-tree.qza

qiime diversity core-metrics-phylogenetic \
  --i-phylogeny rooted-tree.qza \
  --i-table table.qza \
  --p-sampling-depth SELECTED_DEPTH \
  --m-metadata-file metadata.tsv \
  --output-dir core-metrics-results
```

- Choose the sampling depth from the observed library-size distribution and study design; report samples excluded at the selected depth.
- For non-phylogenetic analyses, use the appropriate non-phylogenetic core-metrics workflow. Do not calculate phylogenetic metrics without a suitable tree.

10. Statistical comparisons and reporting

- Select alpha- and beta-diversity metrics before inspecting group outcomes, where possible, and justify them biologically.
- Check group sizes, study design, repeated measures, confounders and multiple-testing corrections before choosing statistical tests.
- Report all quality filtering and denoising parameters, QIIME 2 and plugin versions, reference database version, sampling depth, exclusions, controls and software provenance.
- Retain the .qza and .qzv artifacts, metadata, manifest, command history, provenance and exported figures/tables so that the analysis can be audited.

Quality-control checklist

- FASTQ and metadata sample IDs agree; manifest paths resolve.
- Read-quality plots and trimming/truncation choices are documented.
- DADA2 diagnostics and per-sample read retention are reviewed.
- Negative controls and mock/reference samples are examined, if included in the sequencing design.
- Classifier matches the marker region and reference database version is recorded.
- Rarefaction/sampling-depth decision is justified and its effect on sample retention is reported.

References and authoritative resources

1. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA, Holmes SP. DADA2: High-resolution sample inference from Illumina amplicon data. Nature Methods. 2016;13:581-583. doi:10.1038/nmeth.3869.

2. Bolyen E, Rideout JR, Dillon MR, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nature Biotechnology. 2019;37:852-857. doi:10.1038/s41587-019-0209-9.

3. QIIME 2 Amplicon Distribution documentation, stable documentation and q2-dada2 plugin reference. The current QIIME 2 release and plugin documentation should be checked before running commands: https://amplicon-docs.qiime2.org/en/stable/ and

https://library.qiime2.org/plugins/qiime2/q2-dada2/overview.

4. QIIME 2 current conceptual overview of marker-gene analysis: metadata preparation, FASTQ import, demultiplexing, primer trimming, denoising, taxonomy, phylogeny and diversity: https://amplicon-docs.qiime2.org/en/latest/explanations/conceptual-overview/

Document status: literature-informed educational reference. Confirm current editions, organism-specific applicability, and institutional requirements before use. Prepared 10 November 2026.